Midterm Practice Problems

Econ 502: Advanced Microeconomics

Problem 1: Monopoly Pricing

A monopolist faces the inverse demand curve $P = 200 - 2Q$ and has a constant marginal cost of $MC = 40$ with no fixed costs.

a) Derive the marginal revenue function. Find the profit-maximizing quantity Q_m^*, price P_m^*, and the monopolist's profit.

b) What would the equilibrium price and quantity be under perfect competition (i.e., $P = MC$)? Calculate total surplus (consumer surplus + producer surplus) under perfect competition.

c) Calculate total surplus under monopoly and the deadweight loss from monopoly pricing.

d) Compute the Lerner Index at the monopoly equilibrium. Verify your answer using the inverse elasticity rule.

Problem 2: Third-Degree Price Discrimination

A streaming service has two types of subscribers: adults (A) and students (S). The demand in each market is:

- Adults: $Q_A = 100 - P_A$
- Students: $Q_S = 60 - P_S$

The marginal cost of serving each additional subscriber is constant at $c = 20$.

a) If the firm can price discriminate between the two markets, find the optimal price, quantity, and profit in each market.

b) If the firm must charge a single (uniform) price to all subscribers, find the profit-maximizing price, total quantity, and total profit.

c) Compute the price elasticity of demand and the Lerner index in each market under price discrimination. Which market gets the higher markup? Explain the intuition.

d) Compare total consumer surplus and total welfare (CS + profit) under the two pricing regimes. Is price discrimination welfare-improving here?

Problem 3: Cournot Duopoly

Two firms compete by simultaneously choosing quantities. Market demand is $P = 150 - Q$, where $Q = q_1 + q_2$. Both firms have constant marginal cost $c = 30$ and no fixed costs.

a) Derive each firm's best response (reaction) function.

b) Find the Cournot-Nash equilibrium: each firm's output, the market price, and per-firm profits.

c) Suppose the two firms collude to maximize joint profits and split output equally. Find the collusive quantity per firm, price, and per-firm profit. If firm 2 sticks to the collusive output, what would firm 1 optimally produce? Would firm 1 want to deviate from the agreement?

d) Fill in the following table comparing three market structures (you already have the Cournot and collusion results; compute the competitive outcome where $P = MC$):

	Price	Total Output	Per-Firm Profit
Competition			
Cournot			
Collusion (Monopoly)			

Problem 4: Bertrand Competition

Two firms produce a homogeneous good and compete by simultaneously choosing prices. Consumers buy from the firm with the lower price; if both charge the same price, they split the market equally. Both firms have constant marginal cost c.

a) Suppose both firms set prices $p_1 = p_2 = c$. Show that this is a Nash equilibrium by verifying that neither firm can profitably deviate.

b) Explain why any pair of prices with $p_1 = p_2 > c$ cannot be a Nash equilibrium.

c) Now suppose costs are asymmetric: firm 1 has $MC_1 = 10$ and firm 2 has $MC_2 = 20$. The market demand is $Q = 100 - P$. What is the Nash equilibrium? What is firm 1's profit? (*Hint: Think about what happens if firm 1 prices just below firm 2's cost.*)

d) The result in parts (a) and (b) is known as the *Bertrand paradox*: just two firms are enough to eliminate all market power. Give two assumptions of the Bertrand model that, if relaxed, would allow firms to earn positive profits in equilibrium.
